

Machine Learning Based Analog Circuit Fault Diagnosis
PDF 13 — Chapter 20: Limitations & Future Scope
Honest Project Limitations | Industrial Roadmap: IoT, Edge AI, Digital Twin, Predictive
Maintenance

PART 1 — Understanding the Limitations of Our Project

Chapter 20 — Limitations, Future Scope & Possible Improvements Part 1 — Understanding the Limitations of Our Project Before Starting Congratulations. Until now we have studied everything our project can do. Now comes one of the most important chapters for your viva. Question. Is our project perfect? No. No engineering project is perfect. Every project has limitations. A good engineer does not hide them. Instead, he understands them and suggests improvements. That is exactly what this chapter is about. Professor's Favorite Question Almost every professor asks this. "What are the limitations of your project?" Many students panic. Don't. After this chapter, this question will become one of your strongest answers. What Is a Limitation? A limitation is simply something the current project cannot do or does not do efficiently. It is not a mistake. It is an opportunity for future work.

PROFESSOR ASKS

What is a project limitation?

EXCELLENT ANSWER

A project limitation is a constraint or capability that the current implementation cannot fully address. Identifying limitations helps guide future improvements and research. Limitation 1 Simulated Dataset Question. Where did our data come from? Answer. LTspice. Question. Is LTspice a real circuit? No. It is a simulator. Question. Does LTspice accurately simulate analog circuits? Yes. Very accurately. Question. Still, is it exactly the same as real hardware? No. Why? Because real circuits have Noise Temperature effects Component tolerances Aging Sensor errors Simulation cannot capture everything.

PROFESSOR ASKS

What is the biggest limitation of your dataset?

EXCELLENT ANSWER

The dataset was generated using LTspice simulations. Although the simulator accurately models circuit behavior, it does not fully capture real-world effects such as electrical noise, component tolerances, temperature variations, and hardware imperfections. Excellent. Limitation 2 Offline Prediction Question. Can our dashboard monitor a live circuit continuously? No. The user must first generate the waveform, save it, and upload it. Question. Is that real-time? No. Real-time diagnosis would require continuous data acquisition from actual hardware. Limitation 3 Limited Circuit Types Question. Which circuits does our project support? RC RL RLC Question. Can it immediately diagnose every analog circuit ever made? No. Need additional training data. Need new features. Need new models.

PROFESSOR ASKS

Can your model classify every analog circuit?

EXCELLENT ANSWER

No. The current implementation is trained specifically for the RC, RL, and RLC circuit configurations included in the dataset. Extending it to additional circuit topologies would require new datasets and retraining. Limitation 4 Limited Fault Classes Suppose tomorrow someone introduces a completely new fault. Question. Can our model recognize it? No.

Machine Learning

cannot classify classes it has never seen. Need new training data.

Real Life Analogy

Imagine teaching a child only dogs and cats. Question. Can the child identify a tiger correctly? Probably not. Exactly the same. Limitation 5 Feature Dependence Remember our 17 features. Question. Suppose feature extraction fails. Question. Can the SVM still classify the waveform? No. Because SVM never sees the waveform. It only sees the feature vector.

PROFESSOR ASKS

Why is feature extraction critical in your project?

EXCELLENT ANSWER

The classifier operates only on the extracted numerical feature vector. Therefore, accurate feature extraction is essential for reliable fault classification. Limitation 6 Manual Dataset Generation Question. How was the dataset created? **LTspice** — Simulation. CSV

Feature Extraction

Question. Was this process fully automatic? Not completely. Large industrial systems would require automated data acquisition. Limitation 7 Hardware Validation Question. Did we validate the model using real oscilloscopes and physical circuits? No. That remains future work. This is actually a very good future publication opportunity.

PROFESSOR ASKS

Was the proposed method experimentally validated on physical hardware?

EXCELLENT ANSWER

The current work was validated using LTspice-generated datasets. Experimental validation using physical circuits and measurement instruments remains an important direction for future work. Does This Make The Project Weak? Absolutely not. Question. Why? Because every research paper has limitations. The important thing is that you understand them. Good researchers always write a "Limitations" section.

Engineering Mindset

Notice something. We are not saying The project failed. We are saying The project works within its current scope. That is a professional answer.

Connection to Our Project

Our project currently supports **LTspice** — RC. RL RLC

17 Features

SVM / ANN

Offline Prediction

Everything outside this scope is future work.

COMMON MISTAKES

MISTAKE: [X] A limitation means the project is bad. Wrong.

MISTAKE: [X] Simulation is useless. Wrong. Simulation is the first step of most engineering research.

MISTAKE: [X] Machine Learning can classify unknown faults. Wrong.

MISTAKE: [X] Ignoring limitations makes the project look stronger. Wrong. It makes the presentation less credible.

REVISION NOTES

Remember

the main limitations.

Simulation — Not real hardware.

Offline — Not real-time.

Limited Circuits — RC, RL, RLC only.

Limited Faults — Known classes only.

Feature Extraction — Essential.

Hardware Validation — Future Work.

MEMORY TRICK

Remember this sentence. "Simulation, Scope, Features, Hardware." These four words summarize the major limitations.

KNOWLEDGE CHECK

Level 1

- What is a project limitation?
- Why is LTspice a limitation?
- Can our model perform real-time monitoring?
- Why can't it classify unknown faults?
- Has it been validated on physical hardware?

Level 2

- Explain why simulation and hardware testing are different.
- Why is feature extraction a critical dependency?
- Why can't machine learning recognize unseen fault classes?
- What are the current boundaries of the project?
- Why should limitations be openly discussed?

Professor Level

- Explain the major limitations of your analog fault diagnosis system.
- Why is hardware validation an important next step?
- How would the limitations change if the system were deployed in industry?
- Why is defining the scope of a project important in engineering research?
- How do limitations guide future research directions?
- End of Chapter 20 (Part 1)
- Congratulations.
- You now understand the limitations of your project in a professional and research-oriented manner.
- Instead of fearing questions about limitations, you can now answer them confidently and show that you understand both the strengths and the current boundaries of your work.

Chapter 20 — Future Scope (Part 2)

Including: Real-time fault diagnosis. IoT integration. FPGA implementation. Raspberry Pi deployment. Cloud-based monitoring. Edge AI. Deep Learning improvements. Automatic dataset generation. Industrial deployment. Predictive maintenance. Digital Twin integration. Research paper extensions. M.Tech/PhD continuation ideas. This chapter often impresses professors because it demonstrates that you are thinking beyond the current implementation and understand how the project could evolve into a real industrial system.

PART 2 — Future Scope: Becoming an Industrial Product

Chapter 20 — Limitations, Future Scope & Possible Improvements Part 2 — Future Scope (How This Project Can Become an Industrial Product) Before Starting Every professor asks one question before ending the viva. "If I give you another six months, what improvements will you make?" This is NOT a trick question. They want to know whether you think like a researcher or just a programmer. After this chapter, you'll have one of the strongest answers in your viva. Future Scope 1 Real Hardware Validation Our current project uses LTspice. The next step is building actual RC, RL, and RLC circuits. Use Breadboard PCB Oscilloscope Function Generator Multimeter Collect real waveforms. Train the model again. Now the project becomes industrial-grade.

PROFESSOR ASKS

What is the first future improvement?

EXCELLENT ANSWER

The first future improvement would be validating the proposed methodology using physical hardware instead of relying only on LTspice simulations. Future Scope 2 Real-Time Fault Detection Current workflow looks like **Generate CSV** — Upload. Predict Question. Can the user detect faults while the circuit is running? No. Future system will look like

Live Circuit

Sensor — Microcontroller.

Feature Extraction

Machine Learning

Live Prediction

Now faults are detected immediately. Future Scope 3 IoT Integration Imagine the circuit is 100 km away. Question. Can the engineer still monitor it? Yes. Using IoT. Pipeline becomes **Circuit** — ESP32. **Wi-Fi** — Cloud. **Dashboard** — Prediction. Now the engineer can monitor the circuit from anywhere.

PROFESSOR ASKS

How can IoT improve your project?

EXCELLENT ANSWER

IoT can enable remote transmission of circuit measurements to a cloud server where the machine learning model performs fault diagnosis in real time. Future Scope 4 Raspberry Pi Deployment Question. Should every prediction be performed on a laptop? No. Instead, deploy the trained model on a Raspberry Pi. Now the circuit becomes portable. Useful for field engineers. Future Scope 5 Edge AI Instead of sending data to the cloud, the prediction can happen inside the device. This is called Edge AI. **Advantages** — Faster. Lower latency Better privacy Internet not required Future Scope 6 FPGA Implementation Current project runs using Python. Future version could implement feature extraction or classification using FPGA hardware. **Advantages** — Very high speed. Very low latency Useful for high-frequency applications.

PROFESSOR ASKS

Why use an FPGA?

EXCELLENT ANSWER

FPGA implementation can significantly accelerate feature extraction and classification, making the system suitable for high-speed real-time fault diagnosis. Future Scope 7 Deep Learning Question. Suppose our dataset grows from 5,000 samples to 5 million. Question. Should we still use SVM? Maybe not. Now

Deep Learning

becomes more attractive. Possible models CNN LSTM Transformer Autoencoder Each model could learn more complex relationships. Future Scope 8 Automatic Feature Extraction Currently we manually designed 17 features. Future systems could learn features automatically. Example

Raw Waveform

CNN

Automatic Features

Classifier No manual feature engineering. Future Scope 9 Larger Fault Library Current system supports specific faults. Future version could include

Multiple Open Faults

Multiple Short Faults

Component Aging

Drift Faults

Intermittent Faults

Mixed Faults

Much larger fault coverage. Future Scope 10 Predictive Maintenance Question. Current project answers "What fault exists?" Future systems could answer "When will the fault occur?" This is called Predictive Maintenance. Instead of repairing after failure, maintenance is scheduled before failure. Industries love this.

PROFESSOR ASKS

What is Predictive Maintenance?

EXCELLENT ANSWER

Predictive maintenance uses historical and real-time data to estimate when a component is likely to fail so that maintenance can be performed before an actual breakdown occurs. Future Scope 11 Cloud Dashboard Current dashboard runs locally. Future version could be web-based. Pipeline **Circuit** — Cloud.

Web Dashboard

Any Browser

No installation required. Accessible worldwide. Future Scope 12 Digital Twin One of the hottest research topics. Question. What is a Digital Twin? It is a virtual copy of a physical system. Future workflow

Real Circuit

Digital Twin

Machine Learning

Fault Prediction

Now the virtual circuit updates continuously as the real circuit changes.

PROFESSOR ASKS

What is a Digital Twin?

EXCELLENT ANSWER

A Digital Twin is a virtual representation of a physical system that continuously reflects the real system using sensor data, enabling monitoring, simulation, and predictive diagnostics. Future Scope 13 Explainable AI (XAI) Current system predicts the fault. Future system could also explain why. Example

Predicted Fault

RC_Open **Reason** — High DC Level. Low RMS

Waveform Shape

This makes the system more trustworthy. Future Scope 14 Automatic Retraining Suppose new faults are discovered. Instead of manually training again, future systems could automatically update the dataset and retrain the classifier. Future Scope 15 Industrial Deployment Imagine our system inside

Power Plants

Manufacturing Automotive Aerospace

Consumer Electronics

Continuous health monitoring of analog circuits. Exactly where projects like this become useful.

Future Research

This project can easily be extended into IEEE paper M.Tech thesis PhD research Possible research topics Hybrid AI Physics-Informed AI

Self-Learning Fault Diagnosis

Transfer Learning

Federated Learning

Edge Intelligence

This project has very good research potential.

Complete Future Pipeline

Imagine

Version 3.0

of this project.

Live Circuit

Sensors — ESP32

Edge AI — Cloud

Digital Twin

Predictive Maintenance

Engineer Notification

Now this becomes a complete industrial fault diagnosis platform.

Biggest Lesson

Notice how our project grew. Simulation

Machine Learning

Dashboard — Engineering. **Industry** — Research. That is the journey of good engineering.

COMMON MISTAKES

MISTAKE: [X] Future work means rewriting the project. Wrong.

MISTAKE: [X] Future work must always use Deep Learning. Wrong.

MISTAKE: [X] IoT automatically improves accuracy. Wrong. It improves deployment, not necessarily classification performance.

MISTAKE: [X] Future scope should be unrealistic. Wrong. It should be technically feasible.

REVISION NOTES

Remember

the biggest improvements.

Hardware Validation — Real Circuits.

IoT — Remote Monitoring.

Edge AI — Fast Prediction.

FPGA — High Speed.

Deep Learning — Large Datasets.

Digital Twin — Virtual Circuit.

Predictive Maintenance — Before Failure.

MEMORY TRICK

*Remember this sentence. "Hardware, IoT, Edge AI, Cloud, Digital Twin, Predictive Maintenance."
These six ideas summarize almost the entire future scope.*

KNOWLEDGE CHECK

Level 1

- What is the first future improvement?
- What is Edge AI?
- What is Predictive Maintenance?
- What is a Digital Twin?
- Why use IoT?

Level 2

- How can this project become a real-time monitoring system?
- Why is hardware validation important?
- What advantages does Raspberry Pi deployment provide?
- Why would Deep Learning become more useful for larger datasets?
- How can cloud computing improve the project?

Professor Level

- Explain the complete future roadmap for your project.
- How would you convert this project into an industrial product?
- Discuss the role of IoT, Edge AI, and Cloud Computing in future fault diagnosis systems.
- How could Predictive Maintenance transform analog circuit monitoring?
- Which future extension do you believe has the highest research potential, and why?
- End of Chapter 20
- Congratulations.
- You now have a professional future roadmap for your project.
- You can confidently answer questions about industrial deployment, research extensions, and long-term improvements. This is often one of the sections that leaves a strong impression during project evaluations because it shows you are thinking beyond the current implementation.

Author's Note At this point, the core Project Bible is essentially complete. The remaining chapters I would add are specialized appendices rather than core theory:

Chapter 21: 300+ Viva Questions with model answers. Chapter 22: Placement Interview Questions (SDE, Data Analyst, ML Engineer, Core Electronics). Chapter 23: Complete Python Code Explained Line-by-Line (every function, every variable). Chapter 24: Complete Dashboard Source Code Explained. Chapter 25: Troubleshooting Guide (every common bug and solution). Chapter 26: Project Presentation Script (15-minute, 30-minute, and 45-minute versions). Chapter 27: IEEE Research Paper Style Explanation of the Entire Project. Chapter 28: One-page, five-page, and ten-page revision notes for quick preparation before interviews and vivas. These chapters are not about building the project—they're about mastering it so you can confidently present, defend, and adapt it in any academic or professional setting.